

Chap. 40 The Principle of Statistical Mechanics

1. Effusion of Gas

The escape of gas through a tiny pore or pinhole in its container is called *effusion*. Effusion occurs if the diameter of the hole is considerably smaller than the mean free path of the gas particles. Things that effuse can be either *molecules* or *photons* depending on the constituent of the gas. The effusion of light from the hole of a cavity is a typical kind of black-body radiation.

The effusion rate Γ is defined as the number of particles escaped from the hole per unit of time and per hole area. Let n be the number density of gas particles. You might expect the effusion rate being proportional to the product of n and the mean speed of particle $\bar{v}$.

$$\Gamma \propto n\bar{v}.$$

Because if the particle is to escape from the container, it must first touch the hole area A . Within the time interval dt , the particles can fly for a distance roughly $\bar{v}dt$. So in each time slice dt , the particles in the volume $A\bar{v}dt$ will be flying out of the hole, thus the number of particles that escape is $dN \sim nA\bar{v}dt$, which gives you a rough estimate of the effusion rate $\Gamma = dN/(Adt)$.

But this estimate cannot give us the correct coefficient. To obtain the formula for effusion rate, let us assume the speed of the particles follows a probability distribution described by the function $f(v)$ such that the probability finding a particle with speed between $v \sim v + dv$ is $f(v)dv$. Then the mean speed of the gas particle is well defined as

$$\bar{v} \equiv \int_0^{\infty} dv v f(v).$$

(a) Calculate the effusion rate Γ in terms of n and $\bar{v}$. Hints: Consider particles are flying towards all directions with equal probability. Only those that flies towards the hole will possibly get escape.

Let us consider the effusion of molecules first.

(b) Suppose the mass of the molecule is m , what is the speed distribution function $f(v)$ given the temperature T of the gas? Note: The result is known as Maxwell's speed distribution.

(c) Use the function $f(v)$ to calculate $\bar{v}$, and give the effusion rate as a function of n , m and T . Note: The effusion rate is inversely proportional to the mass of the molecule, thus provide us a method for isotope separation.

Now let us turn to the photon gas. As all photons moves with the same speed, the mean speed for photon is simply the speed of light $\bar{v} = c$. However the density of photon depends on temperature of the cavity.

(d) Find the effusion rate for photon as a function of temperature. Hint: You may find the temperature dependence of the number density of photons in a 3D cavity simply by thermodynamic arguments.

Chap. 42 Applications of Kinetic Theory

1. Polar Dielectrics

There are three kinds of dielectrics: polar, non-polar dielectrics, and ferroelectrics. The molecules of polar dielectric materials have permanent dipole moments, with fixed magnitudes and random orientations. Thus the polar dielectric can be considered as a set of free electric dipoles responding to the external electric field. A free dipole $\mathbf{p}$ in the electric field $\mathbf{E}$ has the potential energy

$$V = -\mathbf{p} \cdot \mathbf{E}.$$

(a) If dipoles obey Boltzmann statistics, what is the thermal average of dipole moment $\langle \mathbf{p} \rangle$ given the temperature T under external electric field $\mathbf{E}$?

If we consider unit volume of substance, we have dipole moment density, i.e. the polarization

$$\mathbf{P} = n\langle \mathbf{p} \rangle,$$

where n is the number density of dipole. In the weak field limit ($E \rightarrow 0$), the polarization of polar dielectric will linearly respond to E , hence

$$\mathbf{P} = \chi \epsilon_0 \mathbf{E}$$

defining the polarizability χ .

(b) Find the polarizability χ in the weak field limit.

The thermal motion randomize the dipole orientations, thus suppressing the polarizability.

2. Thermodynamics Origin of Intermolecular Force

The dipole-dipole interaction, also called Keesom interaction, is a kind of intermolecular force caused by the attraction between permanent dipoles in molecules. Let us suppose two molecules with electric dipole moments $\mathbf{p}_1$ and $\mathbf{p}_2$ respectively separating by the relative displacement $\mathbf{r}$. The magnitude of the dipole moments are the same, $|\mathbf{p}_1| = |\mathbf{p}_2| = p$, but the orientation can be different due to thermal fluctuation.

However, to simplify the calculation, let us project our problem in one dimension. Assuming each dipole only aligns in parallel or anti-parallel to their intermolecular displacement $\mathbf{r}$, which means if we define $p_1 r = \mathbf{p}_1 \cdot \mathbf{r}$, $p_2 r = \mathbf{p}_2 \cdot \mathbf{r}$, the scalars p_1 and p_2 only takes the value of $\pm p$.

(a) Show that the electrostatic potential energy between two dipoles along the same line is given by

$$V = \frac{p_1 p_2}{2\pi \epsilon_0 r^3}.$$

Due to thermal fluctuations, one cannot tell for sure whether p_1 and p_2 are of the same sign or not. However, different configurations may have different Boltzmann weight, and thus physical quantities should be averaged accordingly.

(b) Find the average potential energy $\bar{V}$ of the dipole-dipole system. Is it an attractive interaction or a repulsive one?

(c) Show that $\bar{V} \propto r^{-3}$ at low temperature, and $\bar{V} \propto r^{-6}$ when the temperature is high.

Note: The temperature being considered high or low is by comparison to the energy scale of static dipole-dipole interaction. The intermolecular interaction becomes stronger at low temperature. With strong attractions, molecules will condensed into liquid or solid. As the

temperature rises, the interaction becomes weaker, and molecules evaporate into gas. For molecules in gas, the dipole-dipole interaction no longer follows the r^{-3} law, but shows the r^{-6} behavior. This is a thermodynamic effect.

Chap. 44 The Law of Thermodynamics

1. Finding the Entropy

For equilibrium system, the entropy is a function of state and can be determined up to a constant, from the equation of state and internal energy law. In this problem, we will determine the entropy of gas, solid and light.

All ideal gas follows the equation of state, $pV = Nk_B T$. For gas of molecules, the internal energy is $U = C_V T$, where the constant volume specific heat C_V is considered a constant independent of temperature. For photon gas (light), $U = aVT^4$, where a is also a constant.

- Determine the entropy $S(T, V)$ of molecule gas as a function of temperature and volume.
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(d) Determine the entropy $S(T, V)$ of photon gas as a function of temperature and volume.

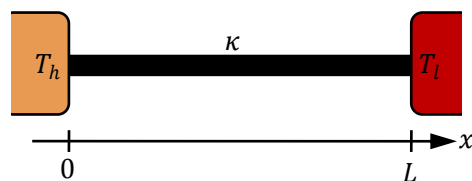
(e) Determine the entropy $S(V, p)$ of photon gas as a function of volume and pressure.

The equation of state for the incompressible idea solid simply states that the volume is a constant $V = \text{const}$. The internal energy can be given by $U = mcT$, where m is the mass of the solid, and c is the specific heat capacity.

(f) Determine the entropy $S(T)$ of incompressible idea solid as a function of temperature.

2. Entropy Production in Heat Conduction

A bar of length L , sectional area A and thermal conductivity κ is in contact with two heat reservoirs of different temperatures T_h and T_l (suppose $T_h > T_l$).



Assume the temperature gradient is uniform throughout the bar.

- Write down the temperature $T(x)$ as a function of position x .
- What is the amount of heat dQ flowing through the sectional area A of the bar in time dt ?
- How much entropy flows into the bar from the high temperature reservoir, and how much entropy flows out of the bar to the low temperature reservoir, during the time dt ?

You will see that there is more entropy flowing out of the bar. The bar gains negative entropy of the heat reservoirs, but its thermodynamic state does not change. The where does this part of entropy go? This is a paradox. Because there is no conservation law for entropy. Even if the state of the system does not change, entropy can be produced by non-equilibrium

process, such as heat transport. The entropy production rate $\sigma = dS_{\text{prod}}/(dt V)$ measures the entropy production per unit of time per unit volume.

(d) Find the entropy production rate $\sigma(x)$ at each point x along the bar. On which side is the entropy production faster, the hotter side or the colder side?

(e) Show that the entropy production along the whole bar exactly balanced the negative entropy from the reservoirs, so that the state of the bar is not changed.

Chap. 45 Illustration of Thermodynamics

1. Thermodynamics of Solid

A often used phenomenological internal energy for solids is

$$U = A e^{b(V-V_0)^2} S^{\frac{4}{3}} e^{\frac{S}{3R}},$$

where A , b and V_0 are positive constants and R is the ideal gas constant; U is the internal energy, S the entropy and V the volume. We will see that many thermodynamic properties of solid can be deduced from this internal energy equation.

(a) Show that this internal energy satisfies the third law of thermodynamics. Hint: You may first find the temperature from the first law of thermodynamics.

(b) Show that the constant volume specific heat C_V is proportional to T^3 at low temperature (Debye's Law), and that it approaches to $3R$ at high temperature (Dulong-Petit's Law).

(c) Calculate the pressure p . What is the physical interpretation of V_0 ?

The thermal expansion coefficient α_p at constant pressure is defined as

$$\alpha_p = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p.$$

(d) Find the behavior of α_p for $p \rightarrow 0$. Is this behavior reasonable?

(e) Show that

$$\left(\frac{\partial p}{\partial T} \right)_V = - \frac{6bRS(4R + S)(V - V_0)}{4R^2 + 8RS + S^2}.$$

Note: The equation of state can be solved from this differential equation, however the result may be complicated.